11.6

2. $f_x = 3y^4x^2 + 4y^3x^3 \quad f_x(1,1) = 7$

$f_y = 4x^3y^3 + 3x^4y^2 \quad f_y(1,1) = 7 \quad \nabla f(1,1) = \begin{pmatrix} 7 \\ 7 \end{pmatrix}$

$\theta = \frac{\pi}{6} \quad \vec{u} = \begin{pmatrix} \cos\frac{\pi}{6} \\ \sin\frac{\pi}{6} \end{pmatrix} \quad \vec{u} = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{pmatrix}$

$D_{\vec{u}} f(1,1) = \begin{pmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 7 \\ 7 \end{pmatrix} = \frac{7}{2}(\sqrt{3}+1)$

4. (a) $f_x = -\frac{y^2}{x^2} \quad f_y = \frac{2y}{x} \quad \nabla f = \left(-\frac{y^2}{x^2}, \frac{2y}{x}\right)$

(b) $\nabla f(1,2) = (-4, 4)$

(c) $D_{\vec{u}} f(1,2) = \begin{pmatrix} -4 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} \frac{2}{3} \\ \frac{\sqrt{5}}{3} \end{pmatrix} = -\frac{8}{3} + \frac{4\sqrt{5}}{3}$

6. $f_x = y^3 z e^{xyz} \quad f_y = xy^2 z e^{xyz} \quad f_z = xy^3 e^{xyz}$

(a) $\nabla f = (y^3 z e^{xyz}, xy^2 z e^{xyz}, xy^3 e^{xyz})$

(b) $\nabla f(0,1,-1) = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$

(c) $D_{\vec{u}} f(0,1,-1) = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{3}{13} \\ \frac{4}{13} \\ \frac{12}{13} \end{pmatrix} = -\frac{3}{13}$

10. $g_r = \frac{s}{1+r^2s^2} \quad g_s = \frac{r}{1+r^2s^2} \quad \nabla g = \left(\frac{s}{1+r^2s^2}, \frac{r}{1+r^2s^2}\right)$

$\nabla g(1,2) = \left(\frac{2}{5}, \frac{1}{5}\right) \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{5i+10j}{\sqrt{5^2+10^2}} = \left(\frac{\sqrt{5}}{5}, 2\frac{\sqrt{5}}{5}\right)$

$D_{\vec{u}} g(1,2) = \nabla g(1,2) \cdot \vec{u} = \begin{pmatrix} \frac{2}{5} \\ \frac{1}{5} \end{pmatrix} \cdot \begin{pmatrix} \frac{\sqrt{5}}{5} \\ 2\frac{\sqrt{5}}{5} \end{pmatrix} = \frac{4\sqrt{5}}{25}$

12. $D_{\vec{u}} f(2,2) = \nabla f(2,2) \cdot \vec{u} = -3$

17. $f_x = \frac{x}{\sqrt{x^2+y^2+z^2}} \quad f_y = \frac{y}{\sqrt{x^2+y^2+z^2}} \quad f_z = \frac{z}{\sqrt{x^2+y^2+z^2}}$

$\nabla f(3,6,-2) = \left(\frac{3}{7}, \frac{6}{7}, -\frac{2}{7}\right) \quad |\nabla f(3,6,-2)| = \sqrt{(\frac{3}{7})^2+(\frac{6}{7})^2+(\frac{2}{7})^2} = 1$

maximum rate $= 1$ direction $= \left(\frac{3}{7}, \frac{6}{7}, -\frac{2}{7}\right)$

19. (a) $|\nabla f(x)|$ is the maximum rate, $\nabla f(x)$ is its direction

$\vec{u} = \frac{\nabla f(x)}{|\nabla f(x)|} \quad D_{-\vec{u}} f(x) = \nabla f(x) \frac{-\nabla f(x)}{|\nabla f(x)|} = -|\nabla f(x)|$

$-|\nabla f(x)|$ is maximum decrease rate

(b) $f_x = 4x^3y - 2xy^3 \quad f_y = x^4 - 3x^2y^2$

$\nabla f(2,-3) = (12, -92) \quad \frac{\nabla f}{|\nabla f|} = \frac{(12,-92)}{\sqrt{12^2+92^2}} = \left(\frac{3}{\sqrt{538}}, -\frac{23}{\sqrt{538}}\right)$

direction $= \left(-\frac{3}{\sqrt{538}}, \frac{23}{\sqrt{538}}\right)$

21. $f_x = 2x-2 \quad f_y = 2y-4$

$\nabla f = (2x-2, 2y-4) \quad \begin{cases} 2x-2 = 2y-4 \\ x > 1 \\ y > 2 \end{cases} \Rightarrow y = x+1$

so $y = x+1$ and $x > 2$ is the points required

23. (a) $T = \frac{m}{\sqrt{x^2+y^2+z^2}} \quad 120 = \frac{m}{\sqrt{1+2^2+2^2}} \Rightarrow m = 360$

$T = \frac{360}{\sqrt{x^2+y^2+z^2}} \quad T_x = \frac{360x}{(x^2+y^2+z^2)^{\frac{3}{2}}} \quad T_y = \frac{360y}{(x^2+y^2+z^2)^{\frac{3}{2}}} \quad T_z = \frac{360z}{(x^2+y^2+z^2)^{\frac{3}{2}}}$

$\nabla T(1,2,2) = \left(\frac{40}{3}, \frac{80}{3}, \frac{80}{3}\right) \quad \vec{v} = (1,-1,1) \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \left(\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$

$D_{\vec{u}} T(1,2,2) = \nabla T(1,2,2) \cdot \vec{u} = \left(\frac{40}{3}, \frac{80}{3}, \frac{80}{3}\right)\left(\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) = \frac{40\sqrt{3}}{9}$

(b) $\nabla T(x,y,z) = \left(\frac{360x}{(x^2+y^2+z^2)^{\frac{3}{2}}}, \frac{360y}{(x^2+y^2+z^2)^{\frac{3}{2}}}, \frac{360z}{(x^2+y^2+z^2)^{\frac{3}{2}}}\right)$

∇T is the maximum increase rate, the direction

is given by the vector towards the origin

26. (a) $z_y = -0.02y \quad z_y(60,40) = -0.8 \quad \vec{u} = (0,-1)$

$D_{\vec{u}} z = (-0.6, -0.8) \cdot (0,-1) = 0.8 > 0$ ascend

(b) $\vec{v} = (-1, 1) \quad \vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{(-1,1)}{\sqrt{1^2+1^2}} = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

$D_{\vec{u}} z = (-0.6, -0.8) \cdot \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = -\frac{\sqrt{2}}{10}$ descend rate: $-\frac{\sqrt{2}}{10}$

(c) $\nabla z(60,40) = (-0.6, -0.8) \quad \vec{u} = \frac{(-0.6,-0.8)}{\sqrt{0.6^2+0.8^2}} = (-0.6, -0.8)$

direction: $(-0.6, -0.8)$; rate: 1; $\arctan(1) = 45°$

29. (a) let $f = au + bv \quad u(x,y) \quad v(x,y) \quad \nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) \quad \nabla v = \left(\frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}\right)$

$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = \left(\frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x}, \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y}\right)$

$= \left(a\frac{\partial u}{\partial x} + b\frac{\partial v}{\partial x}, a\frac{\partial u}{\partial y} + b\frac{\partial v}{\partial y}\right)$

$= a\nabla u + b\nabla v$

(b) $\nabla(uv) = \left(\frac{\partial(uv)}{\partial x}, \frac{\partial(uv)}{\partial y}\right) = \left(\frac{v\partial u}{\partial x} + \frac{u\partial v}{\partial x}, \frac{v\partial u}{\partial y} + \frac{u\partial v}{\partial y}\right)$

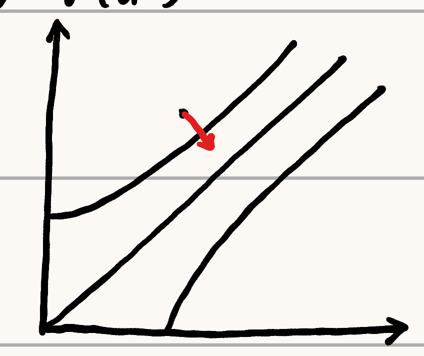
$= u\nabla v + v\nabla u$

(c) $\nabla\left(\frac{u}{v}\right) = \left(\frac{\partial(\frac{u}{v})}{\partial x}, \frac{\partial(\frac{u}{v})}{\partial y}\right) = \left(\frac{v\partial u}{v^2\partial x} - \frac{u\partial v}{v^2\partial x}, \frac{v\partial u}{v^2\partial y} - \frac{u\partial v}{v^2\partial y}\right)$

$= \frac{v\nabla u - u\nabla v}{v^2}$

(d) $\nabla(u^n) = \left(\frac{\partial(u^n)}{\partial x}, \frac{\partial(u^n)}{\partial y}\right) = \left(\frac{nu^{n-1}\partial u}{\partial x}, \frac{nu^{n-1}\partial u}{\partial y}\right) = nu^{n-1}\nabla u$

30.



the gradient is perpendicular to the level curve, the gradient is similar to one unit increasing rate is quite similar to one.

34. (a) $f = xy + yz + zx - 5 \quad \nabla f = (y+z, x+z, x+y)$

$\nabla f(1,2,1) = (3,2,3) \quad P: 3x + 2y + 3z = 10$

(b) $\frac{x-1}{3} = \frac{y-2}{2} = \frac{z-1}{3}$

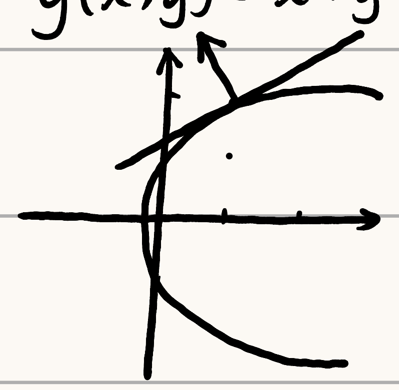
36. (a) $f = x^4 + y^4 + z^4 - 3x^2y^2z^2 \quad \nabla f = (4x^3 - 6xy^2z^2, 4y^3 - 6x^2yz^2, 4z^3 - 6x^2y^2z)$

$\nabla f(1,1,1) = (-2,-2,-2) \quad P: -2x - 2y - 2z = -6$

(b) $\frac{x-1}{-2} = \frac{y-1}{-2} = \frac{z-1}{-2} \Rightarrow x = y = z$

40. $\nabla g = (2x-4, 2y) \quad \nabla g(1,2) = (-2,4)$

$g(x,y) = x^2 + y^2 - 4x = 1 \Rightarrow (x-2)^2 + y^2 = 5$



$k = \frac{1}{2} \quad y = \frac{1}{2}(x-1) + 2 \Rightarrow y = \frac{x}{2} + \frac{3}{2}$

41. $f = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \quad \nabla f = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2}\right)$

$\nabla f(x_0, y_0, z_0) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2}\right)$

$\therefore$ tangent plane $\frac{2x_0}{a^2}(x - x_0) + \frac{2y_0}{b^2}(y - y_0) + \frac{2z_0}{c^2}(z - z_0) \Rightarrow$

$\frac{x_0}{a^2}x + \frac{y_0}{b^2}y + \frac{z_0}{c^2}z = 1$

43. $\nabla f = (2x, -2y, -2z)$

$\vec{n} = (1,1,-1) \quad \frac{2x}{1} = \frac{-2y}{1} = \frac{-2z}{-1} \Rightarrow \begin{cases} y = -x \\ z = x \end{cases}$

$x^2 - x^2 - x^2 = 1 \Rightarrow -x^2 = 1 \quad \therefore$ no

45. $f = x^2 + y^2 - z \quad \nabla f = (2x, 2y, -1) \quad \nabla f(1,1,2) = (2,2,-1)$

$\frac{x-1}{2} = \frac{y-1}{2} = \frac{z-2}{-1} \Rightarrow \begin{cases} y = x \\ z = -\frac{x}{2} + \frac{5}{2} \end{cases}$

$-\frac{x}{2} + \frac{5}{2} = 2x^2 \Rightarrow 4x^2 + x - 5 = 0 \Rightarrow \begin{cases} x = 1 \\ x = -\frac{5}{4} \end{cases} \quad \left(-\frac{5}{4}, -\frac{5}{4}, \frac{15}{8}\right)$

49. $F_1: z = x^2 + y^2 \quad \nabla F_1 = (2x, 2y, -1) \quad \nabla F_1(-1,1,2) = (-2,2,-1)$

$F_2: 4x^2 + y^2 + z^2 = 9 \quad \nabla F_2 = (8x, 2y, 2z) \quad \nabla F_2(-1,1,2) = (-8,2,4)$

$\nabla F_1 \times \frac{\Delta F_2}{2} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} -4 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ 6 \end{pmatrix}$

tangent line: $\frac{x+1}{5} = \frac{y-1}{8} = \frac{z-2}{6} \Rightarrow x = -1 + 5t \quad y = 1 + 8t \quad z = 2 + 6t$

52. (a) for any $\varepsilon > 0 \quad \delta = \sqrt[3]{\varepsilon^2}$ when $\sqrt{x^2+y^2} < \delta \quad \sqrt[3]{|xy|} \leq \sqrt[3]{x^2+y^2} < \delta^{\frac{2}{3}} = \varepsilon$

$\therefore \lim_{(x,y)\to(0,0)} \sqrt[3]{xy} = 0 = f(0,0) \quad \therefore f(x,y)$ is continuous

$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \frac{0-0}{h} = 0 \quad \therefore f_x(0,0) = 0$

similarly $f_y(0,0) = 0$

$D_{\vec{u}} f(0,0) = \lim_{h\to 0^+} \frac{f(0+hu_1, 0+hu_2) - f(0,0)}{h} = \lim_{h\to 0^+} \frac{\sqrt{|hu_1 hu_2|}}{h} = \frac{|h|\sqrt{u_1 u_2}}{h}$

when $h > 0 \quad D_{\vec{u}} f(0,0) = \sqrt{u_1 u_2} \quad h < 0 \quad D_{\vec{u}} f(0,0) = -\sqrt{u_1 u_2}$

because $\sqrt{u_1 u_2}$ is required to equal to $-\sqrt{u_1 u_2} \quad \therefore u_1 \cdot u_2 = 0$

$u = (0,1)$ or $(1,0) \quad \therefore$ only x, y axis exist the directional deriatives

53. $\vec{u} = (u_1, u_2) \quad \vec{v} = (v_1, v_2) \quad D_{\vec{u}} f = \nabla f \cdot \vec{u} = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$

$D_{\vec{v}} f = \nabla f \cdot \vec{v} = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{pmatrix} \cdot \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \quad \left(\begin{array}{cc|c} u_1 & u_2 & D_{\vec{u}} f \\ v_1 & v_2 & D_{\vec{v}} f \end{array}\right)$ can calculate ∇f